

Magnetic Effects of Electric Current (Set A) — MCQ Worksheet

Science • Chapter 12 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. The relationship between electricity and magnetism was first demonstrated by:

- (A) Faraday
- (B) Oersted
- (C) Ampere
- (D) Maxwell

Q2. Magnetic field lines around a bar magnet emerge from the:

- (A) S pole
- (B) N pole
- (C) Both poles
- (D) Neither pole

Q3. Magnetic field lines never:

- (A) Form loops
- (B) Cross each other
- (C) Touch poles
- (D) End

Q4. The closeness of magnetic field lines indicates:

- (A) Weak field
- (B) Strong field
- (C) No field
- (D) Reversed field

Q5. The magnetic field around a straight current-carrying conductor consists of:

- (A) Straight lines
- (B) Concentric circles
- (C) Parabolas
- (D) Random shapes

Q6. The right-hand thumb rule gives the direction of:

- (A) Force
- (B) Magnetic field around a wire
- (C) Electric field
- (D) Current only

Q7. Inside a current-carrying solenoid the magnetic field is:

- (A) Zero
- (B) Uniform and strong
- (C) Weak
- (D) Random

Q8. A solenoid behaves like a:

- (A) Bar magnet
- (B) Single charge
- (C) Capacitor
- (D) Resistor

Q9. An electromagnet is made stronger by inserting a:

- (A) Wooden core
- (B) Plastic rod
- (C) Soft iron core
- (D) Glass rod

Q10. Fleming's left-hand rule gives the direction of:

- (A) Induced current
- (B) Magnetic field
- (C) Force on a current-carrying conductor in a magnetic field
- (D) Magnetic flux

Q11. In Fleming's left-hand rule, the forefinger represents the:

- (A) Force
- (B) Current
- (C) Magnetic field
- (D) Velocity

Q12. An electric motor converts:

- (A) Mechanical to electrical energy
- (B) Electrical to chemical energy
- (C) Electrical to mechanical energy
- (D) Light to electrical energy

Q13. In a DC motor, the device used to reverse current direction in the coil every half rotation is the:

- (A) Brushes
- (B) Split-ring commutator
- (C) Battery
- (D) Switch

Q14. Electromagnetic induction was discovered by:

- (A) Oersted
- (B) Maxwell
- (C) Faraday
- (D) Edison

Q15. An electric generator works on the principle of:

- (A) Heating effect
- (B) Magnetic effect of current
- (C) Electromagnetic induction
- (D) Chemical effect

Q16. AC current in India has a frequency of:

- (A) 10 Hz
- (B) 50 Hz
- (C) 60 Hz
- (D) 100 Hz

Q17. Fleming's right-hand rule gives the direction of:

- (A) Force on a conductor
- (B) Induced current
- (C) Magnetic field around a wire
- (D) Velocity

Q18. In a domestic 3-wire system, the neutral wire is normally at:

- (A) 220 V
- (B) 0 V
- (C) 110 V
- (D) Negative voltage

Q19. The colour code for the earth wire is generally:

- (A) Red
- (B) Black
- (C) Green
- (D) Yellow

Q20. Short-circuit occurs when:

- (A) Live and neutral wires touch directly
- (B) Earth wire is loose
- (C) Voltage falls
- (D) Fuse blows

Q21. The principle behind the working of an electromagnet is the:

- (A) Heating effect of current
- (B) Magnetic effect of current
- (C) Chemical effect
- (D) Light effect

Q22. A current-carrying conductor experiences maximum force in a magnetic field when current is:

- (A) Parallel to field
- (B) Anti-parallel to field
- (C) Perpendicular to field
- (D) At 30° to field

Q23. Galvanometer detects:

- (A) Voltage only
- (B) Resistance only
- (C) Presence of small currents
- (D) Magnetic field strength only

Q24. A coil rotated in a magnetic field generates AC because:

- (A) Field reverses
- (B) Direction of induced EMF reverses every half rotation
- (C) Coil resists
- (D) Magnets repel

Q25. An MCB (Miniature Circuit Breaker) acts like a:

- (A) Voltage stabiliser
- (B) Fuse with reusable mechanism
- (C) Battery
- (D) Capacitor